Vibration and Damping Analysis of a Bridge Stay Cable

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Abstract— **This final report presents an extended 2D dynamic model for the vibration analysis of a stay cable subjected to gravity, aerodynamic loading, mechanical damping, and deck-induced excitation. Building upon the midterm framework, the final implementation incorporates: (i) full time-resolved visualization of cable displacements, internal and external forces, and modal behavior; (ii) prescribed deck vibration as a moving boundary condition; (iii) comprehensive energy-based diagnostics; and (iv) frequency-domain analysis using FFT to identify dominant vibration modes. The cable is modeled as a lumped-mass chain with axial stretching and bending stiffness and is solved using an implicit Newton–Raphson time integrator suitable for stiff, nonlinear systems. Simulation results illustrate the combined influence of aerodynamic drag, lift, viscous damping, and deck motion on cable oscillations. These findings provide a robust computational foundation for studying wind-induced vibration mechanisms and evaluating the dynamic impact of deck excitation on stay cables.**

I. Problem Statement and Motivation

Stay cables are key load-carrying components of modern cable-stayed bridges. Because of their long, slender, and low inherent damping, they are prone to a variety of wind and rain-induced vibration mechanisms, including rain-wind-induced vibration and parametric excitation driven by deck motion [1]. These vibrations can lead to many problems, like fatigue damage and local component failure, if not properly controlled.

To better understand the interaction between wind loading, gravity, cable tension, and damping, this project develops a simplified numerical model of a single stay cable. The current version of the code focuses on a two-dimensional plane model. The cable is modeled as a sequence of lumped nodes connected by stretching and bending elements, and is subjected to gravity, steady uniform wind, aerodynamic drag and lift, and local mechanical damping.

The motivation for this work is twofold. The first objective is to establish a computational framework capable of resolving the time-varying dynamic response of a stay cable under realistic wind and damping conditions through a robust implicit time-integration scheme suited for stiff cable systems. The second objective is to provide a foundation for future extensions, including time-dependent boundary conditions such as deck vibration, damping devices, and comprehensive vibration analyses in both time and frequency domains.

II. Physical Model and Assumptions

The cable examined in this project represents a typical long-span stay cable of approximately 300 m in length [2]. It is assumed to lie within a vertical plane and is anchored between a lower deck point and an upper tower point. For numerical discretization, the cable is represented by $n_v = 51$ lumped-mass nodes connected by $n_e = n_v - 1 = 50$ edges of equal reference length $\Delta l = 6\,\mathrm{m}$, resulting in $n_{\mathrm{dof}} = 2n_v$ degrees of freedom. The cable spans $L = 300\,\mathrm{m}$ with an inclination angle of $\theta = 23^\circ$. The fixed degrees of freedom are the two end nodes:

$$(x_0, y_0), \qquad (x_{n_v-1}, y_{n_v-1}).$$

Each edge acts as an axial stretching element, while bending stiffness is introduced through a discrete curvature formulation applied to neighboring triplets of nodes, consistent with standard slender-structure rod and beam models. The cable's cross-section is modeled as a solid circular steel rod of radius $r_0 = 0.060\,\mathrm{m}$, giving axial stiffness EA and bending stiffness EI based on steel material properties:

$$E = 200\,\mathrm{GPa}, \qquad \nu = 0.3, \qquad \rho = 7850\,\mathrm{kg/m^3}.$$

The initial geometry is defined as a straight inclined line between the two supports. A static equilibrium solve—including gravity and axial elasticity—is performed to obtain a sagged configuration with nonzero static tension. This equilibrium state is essential because it governs the natural frequencies, mode shapes, and global dynamic behavior of the cable.

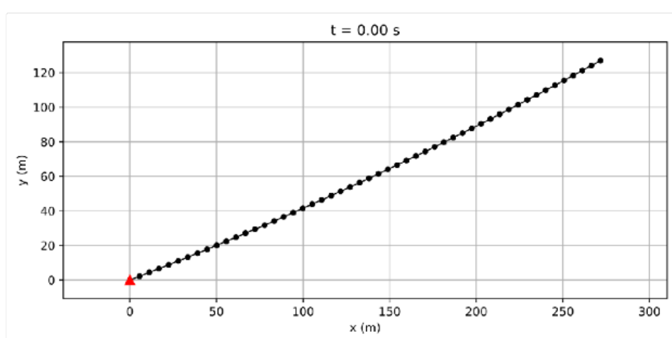

Fig. 1. Initial undeformed configuration of the discretized stay cable

III. Numerical Formulation

The generalized coordinates are collected in a vector $\mathbf{q} \in \mathbb{R}^{2n_v}$, containing the x- and y-coordinates of each node:

$$\mathbf{q} = \left[x_0, y_0, x_1, y_1, \ldots, x_{n_v-1}, y_{n_v-1}\right]^T.$$

Velocities are defined as $\mathbf{u} = \dot{\mathbf{q}}$. For node k, its position and velocity at time step n are denoted by $\mathbf{x}_k^n$ and $\mathbf{u}_k^n$, respectively.

A. Stretching energy

Each edge connects two adjacent nodes, denoted node0 and node1 in the code. The current edge vector is

$$\mathbf{e} = \mathbf{node1} - \mathbf{node0}, \qquad L = \|\mathbf{e}\|,$$

where L is the current length, and l_k is the reference (initial) length of the same edge. The stretching strain is approximated as

$$\varepsilon_x = \frac{L}{l_k} - 1.$$

The stretching energy of the edge is

$$E_s = \frac{1}{2} EA\, l_k\, \varepsilon_x^2,$$

where EA is the axial stiffness (Young's modulus E times the cross-sectional area A). The code computes the gradient $\partial E_s / \partial \mathbf{q}$ and the Hessian $\partial^2 E_s / \partial \mathbf{q}^2$ analytically for each edge in 2D, and assembles them into the global stretching force vector $\mathbf{F}_s$ and Jacobian $\mathbf{J}_s$.

B. Bending energy

Bending resistance is introduced via a discrete curvature κ at interior nodes, based on three consecutive nodes—node0, node1, and node2. The local curvature is obtained from the turning angle between adjacent segments divided by an average segment length. The bending energy at an interior node is

$$E_b = \frac{1}{2} EI\, l_v\, (\kappa - \bar{\kappa})^2,$$

where EI is the bending stiffness, l_v is a Voronoi-type nodal length, and $\bar{\kappa}$ is the natural curvature (zero for a straight initial cable). The code evaluates the gradient and Hessian of E_b numerically using finite differences, yielding the bending force vector $\mathbf{F}_b$ and Jacobian $\mathbf{J}_b$ acting on the three involved nodes.

C. Mass matrix, gravity, and initial tension

Each node carries a lumped mass

$$m = \rho A\, \delta L,$$

where ρ is the material density, A is the cross-sectional area, and δL is the segment length. The global mass matrix $\mathbf{M}$ is diagonal, with identical mass entries for the x and y directions of each node.

Gravity is applied as a constant distributed load along the cable, leading to a nodal force vector $\mathbf{F}_g$ with vertical components proportional to the tributary length associated with each node.

To obtain a realistic initial configuration, a static equilibrium solve is performed under gravity and axial stiffness, with both ends fixed. The result is a sagged cable geometry and a consistent internal tension distribution that serves as the starting point for the dynamic simulation.

D. High-Reynolds-number drag

For each node k, the code computes a backward Euler estimate of the nodal velocity,

$$\mathbf{u}_k = \frac{\mathbf{x}_k^{n+1} - \mathbf{x}_k^n}{\Delta t},$$

where $\mathbf{x}_k^{n+1}$ and $\mathbf{x}_k^n$ are the updated and previous nodal positions. The incoming wind is represented by a uniform background flow velocity

$$\mathbf{u}_f = (30.0, 0.0)\,\mathrm{m/s},$$

corresponding to horizontal wind of 30 m/s.

The relative velocity between the cable and the wind field is therefore

$$\mathbf{u}_{\mathrm{rel},k} = \mathbf{u}_k - \mathbf{u}_f.$$

The aerodynamic drag force acting on node k is modeled using the standard quadratic high–Reynolds-number cross-flow formula for circular cylinders:

$$\mathbf{F}_{\mathrm{drag},k} = -\frac{1}{2}\rho_f\, C_D A_k\, \|\mathbf{u}_{\mathrm{rel},k}\|\, \mathbf{u}_{\mathrm{rel},k},$$

where the fluid density is

$$\rho_f = 1.225\ \mathrm{kg/m^3},$$

and the drag coefficient for the cable is

$$C_D = 0.79.$$

The effective projected area associated with node k is

$$A_k = 2\, r_0\, l_{v,k},$$

where $r_0 = 0.060$ m is the cable radius and $l_{v,k}$ is the Voronoi length surrounding node k. This form is consistent with classical high–Reynolds-number drag models for slender cylindrical structures.

The Jacobian of $\mathbf{F}_{\mathrm{drag},k}$ with respect to $\mathbf{x}_k^{n+1}$ is also computed analytically to construct the drag contribution to the global Jacobian matrix $\mathbf{J}_h$, ensuring stable and fully implicit time integration.

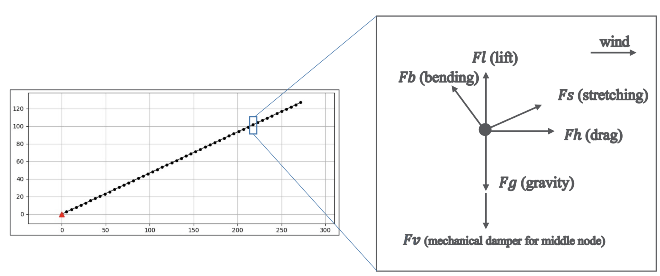

Fig. 2. Forces acting on a cable node

E. Mechanical viscous damper

In addition to fluid-related damping, the model includes a viscous mechanical damper attached to the cable. The damper is applied at the mid-span node, located at index $i = n_v/2$, and acts in the vertical direction. The mechanical damping coefficient used in this study is

$$c_{\text{damp}} = 1000 \text{ N}\cdot\text{s/m}.$$

The damper is represented by a diagonal damping matrix $\mathbf{C}$ acting on the nodal velocities:

$$\mathbf{F}_{\text{v}} = -\mathbf{C}\,\mathbf{u},$$

where the nodal velocity is estimated using a backward Euler update,

$$\mathbf{u} = \frac{\mathbf{q}^{n+1} - \mathbf{q}^{n}}{\Delta t}.$$

In the present implementation, only the vertical degree of freedom of the mid-span node is assigned the non-zero coefficient c_{damp}, reflecting a vertical viscous device anchored to the deck or tower. This mechanism provides an additional energy dissipation pathway and directly increases the effective damping of the cable's fundamental vertical vibration mode.

F. Lift force model

A preliminary lift model is also implemented, analogous to the drag expression. For each node, the aerodynamic lift force is taken as

$$\mathbf{F}_{\text{lift},k} = -\frac{1}{2}\rho_f C_L A_k \|\mathbf{u}_{\text{rel},k}\|^2$$

where C_L is a lift coefficient, A_k is the same projected area used for drag, and the flow direction is to the incoming wind [3]. The lift coefficient used in the lift model is

$$C_L = 0.3.$$

Altogether, the total non-gravitational external force on the cable at each time step is composed of the aerodynamic drag and lift, and the mechanical damper reaction. The dynamic response of stay cables is strongly governed by aerodynamic loading and structural damping. Together, these components enable the simulation framework to capture the essential fluid–structure interactions and energy dissipation pathways that dominate the vibration behavior of long-span cable systems.

G. Total Energy Calculation

The total mechanical energy of the cable system is composed of kinetic energy, potential energy, and cumulative energy dissipation due to mechanical damping and aerodynamic drag. The kinetic energy is computed as

$$E_{\text{k}} = \frac{1}{2}\sum_i m_i \|\mathbf{u}_i\|^2 = \frac{1}{2}\mathbf{u}^{\mathsf{T}} M\,\mathbf{u},$$

where $\mathbf{u}$ is the nodal velocity vector and M is the diagonal mass matrix.

Energy dissipation arises from two mechanisms. Mechanical damping contributes instantaneous power

$$P_{\text{damp}} = \mathbf{u}^{\mathsf{T}} C\,\mathbf{u},$$

and the cumulative dissipated energy is

$$E_{\text{damp}}(t) = \int_0^t P_{\text{damp}}(\tau)\,d\tau.$$

Similarly, the power removed from the system due to high–Reynolds-number aerodynamic drag is

$$P_{\text{drag}} = -\mathbf{F}_h^{\mathsf{T}}\,\mathbf{u},$$

with cumulative drag dissipation

$$E_{\text{drag}}(t) = \int_0^t P_{\text{drag}}(\tau)\,d\tau.$$

The total mechanical energy of the system is therefore expressed as

$$E_{\text{system}}(t) = E_{\text{p}}(t) + E_{\text{k}}(t) - E_{\text{damp}}(t) - E_{\text{drag}}(t),$$

where the total potential energy

$$E_{\text{p}} = E_{\text{stretching}} + E_{\text{bending}} + E_{\text{gravity}}$$

is composed of stretching, bending, and gravitational contributions. This energy formulation enables monitoring of how mechanical damping and aerodynamic forces remove energy from the system and thereby influence cable vibration decay.

IV. Results

The time step is chosen as $\Delta t = 0.01$ s, and simulations are typically run for 10 s of physical time. The code generates snapshots of the cable geometry every few time steps, which can be displayed interactively or saved as figures for later inspection.

A. Construction and Tension Layout

Preliminary simulations indicate that the cable shape remains close to its initial sagged profile, as expected for a stay cable with significant initial tension. Small oscillations superposed on this mean shape are driven by the aerodynamic loading and are progressively damped by both fluid and mechanical damping mechanisms. When the axial stress along the cable is reconstructed from the stretching strain in each edge, it is observed that stress tends to increase from the mid-span toward the anchorage, with the maximum stress appearing near the fixed end. This behavior is consistent with the fact that the anchored end must carry the cumulative tension from the entire cable and is qualitatively in line with engineering observations of high-demand regions in stay cables.

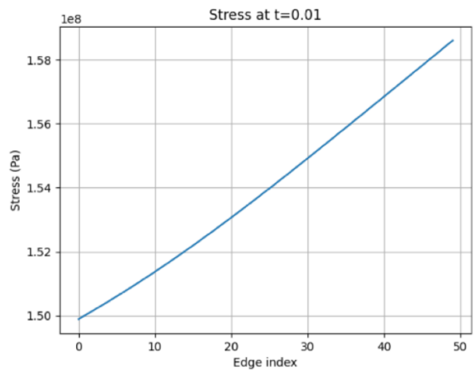

Fig. 3. Increasing tension toward the anchor

B. Energy Evolution and Dissipation Characteristics

The evolution of potential, kinetic, and dissipated energy for the case $C = 200$ illustrates the characteristic exchange and decay mechanisms of a lightly damped stay cable. The potential energy exhibits a slowly decaying oscillatory pattern superposed on a large static baseline determined by the sag and axial tension of the cable. These oscillations correspond to periodic conversion between stretching energy and kinetic energy as the cable vibrates.

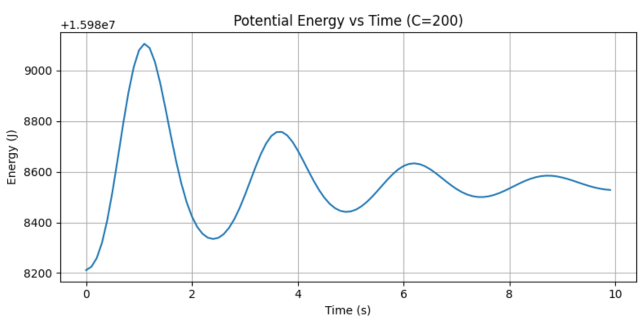

Fig. 4. Potential energy of the cable as a function of time with $c = 200\,\mathrm{N}\cdot\mathrm{s/m}$

The kinetic energy peaks shortly after the initial disturbance and then decays rapidly, reflecting the combined action of mechanical damping at mid-span and aerodynamic drag along the cable. As the vibration amplitude decreases, the kinetic energy oscillations diminish to near zero, indicating that the cable is approaching a quasi-static state.

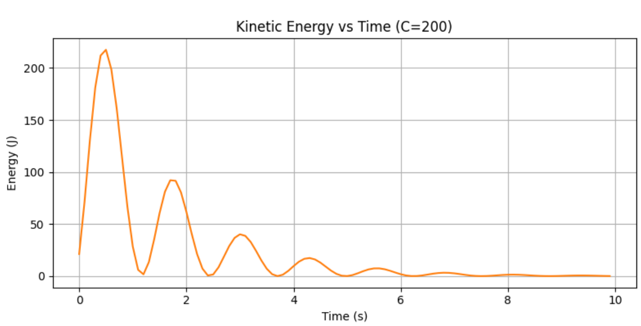

Fig. 5. Kinetic energy of the cable as a function of time with $c = 200\,\mathrm{N}\cdot\mathrm{s/m}$

The dissipated energy, defined as the cumulative sum of mechanical and aerodynamic dissipation, increases monotonically in time but also exhibits oscillatory increments since dissipation is largest when the cable motion is fastest.

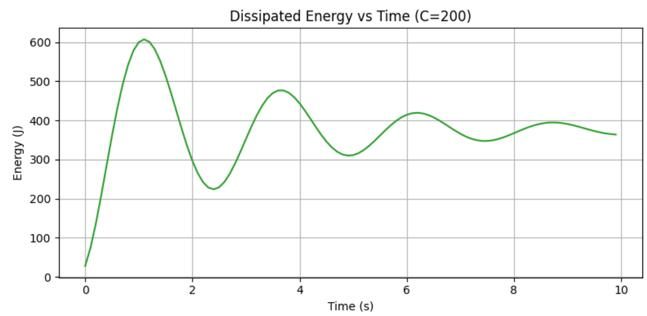

Fig. 6. Dissipated energy of the cable as a function of time with $c = 200\,\mathrm{N}\cdot\mathrm{s/m}$

The decreasing peak amplitudes in all three energy components confirm that the system is losing mechanical energy over time, consistent with expectations for a damped tensioned cable subject to quadratic drag and localized viscous damping. Together, these results validate the correctness of the energy bookkeeping and demonstrate physically meaningful damping behavior.

A comparison of the kinetic energy histories for $C = 200$ and $C = 3000$ demonstrates the strong influence of mechanical damping on vibration decay. For the case $C = 200$, the cable exhibits several cycles of oscillatory motion with gradually decreasing amplitude. The kinetic energy remains for approximately six to seven seconds, indicating that the damping level is relatively weak. In contrast, for the larger damping value $C = 3000$, the decay of kinetic energy is faster. After the initial peak, the oscillation amplitude drops rapidly, and the kinetic energy becomes negligible by $t \approx 3$–4 seconds. These results confirm that increasing the viscous damping coefficient produces a more rapid attenuation of cable vibrations and reduces the duration of transient oscillations, pushing the system toward its quasi-static equilibrium state much more quickly.

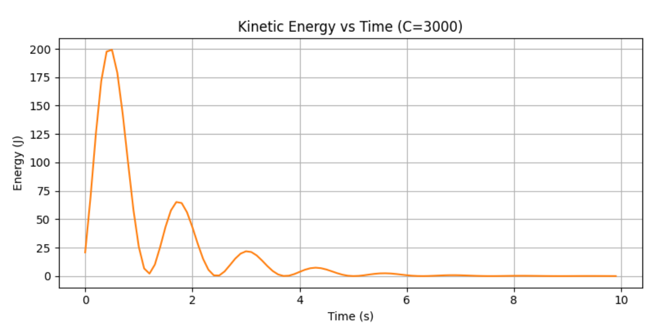

Fig. 7. Dissipated energy of the cable as a function of time with $c = 3000\,\mathrm{N}\cdot\mathrm{s/m}$

C. Vibration Visualization

The time-domain response of the cable was examined by tracking the vertical displacement at midspan under different mechanical damping configurations. When the damping coefficient is small, the cable exhibits slow decay of oscillations, indicating that only a limited amount of vibrational energy is removed at each cycle. As the damping value increases, the amplitude decreases more rapidly and the motion approaches a steady equilibrium in a shorter time [6]. For a damping coefficient around 6000 N·s/m, the system shows very fast attenuation of vibrations and settles into a nearly static configuration, which is consistent with the expected behavior of a lightly disturbed, tension-dominated cable.

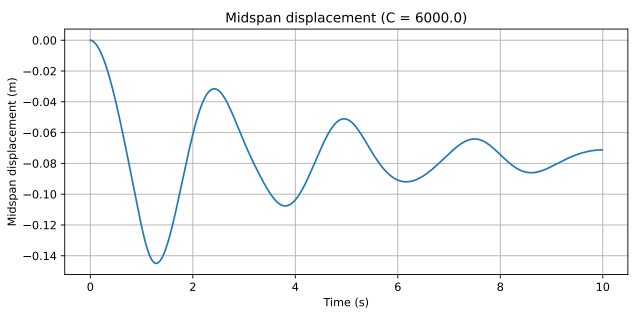

Fig. 8. Midspan displacement response with $c = 6000\,\mathrm{N \cdot s/m}$

A space–time contour visualization further confirms the stability of the system. Because no large external excitation is applied, the cable shape remains almost unchanged throughout the simulation interval. Only small residual oscillations are observed around the static equilibrium. In a system with stronger initial perturbations, clear traveling waves would propagate along the span, but the nearly uniform color distribution in the current contour plot demonstrates that the cable experiences minimal transient motion. Together, the midspan displacement history and the space–time visualization provide a complete picture of how damping controls vibration decay and ensures structural stability.

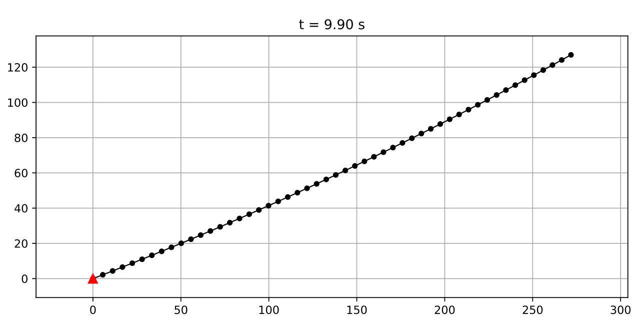

Fig. 9. Cable deformation at $t = 9.9\,\mathrm{s}$

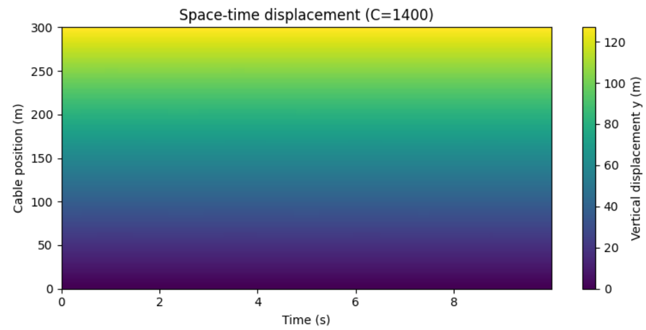

Fig. 10. Space-time displacement plot

D. Critical Damping and Recommended Damper Value

To evaluate the role of mechanical damping, the relationship among the natural frequency, critical damping, and the recommended damper value was examined. The natural frequency represents the oscillation rate the cable would follow in the absence of damping and external forces. When damping is introduced, the system vibrates at a slightly lower value known as the *damped natural frequency*, which governs how the cable behaves under real conditions.

The critical damping coefficient is defined as

$$c_{\text{crit}} = m_{\text{eff}}\, \omega_n,$$

where ω_n is the natural angular frequency and m_{eff} is the effective modal mass associated with the first mode. For long stay cables, engineering literature commonly adopts $m_{\text{eff}} = 0.5M_{\text{total}}$, reflecting that only a portion of the full mass participates significantly in the first-mode vibration. Using a target damping ratio of 2%, the recommended mechanical damping is computed as

$$c_{\text{rec}} = \zeta\, c_{\text{crit}},$$

as discussed in [5].

While this recommended value provides a practical engineering guideline, numerical simulations show that larger damping values suppress vibrations more rapidly. However, the "best" damper is not simply the one that produces the fastest decay. Excessive damping may amplify the transfer of deck motion into the cable, reduce system realism, and introduce unnecessary mechanical demands on the anchorage. Therefore, the recommended damping value represents a balanced design choice, consistent with common practice in cable-stayed bridge vibration control [7].

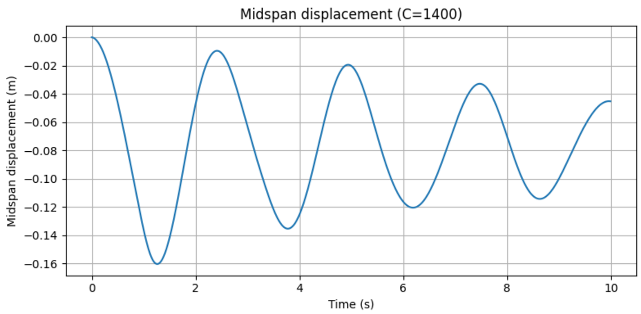

Fig. 11. Midspan displacement response with $c = 1400\,\mathrm{N \cdot s/m}$

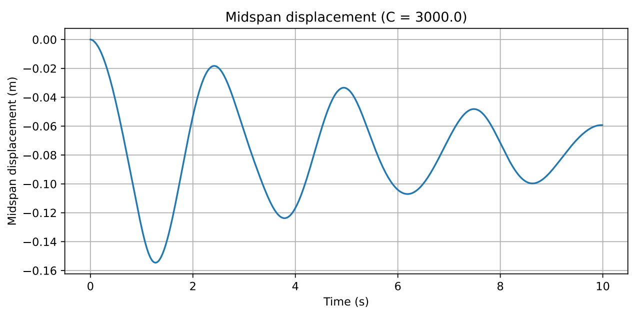

Fig. 12. Midspan displacement response with $c = 3000\,\mathrm{N \cdot s/m}$

E. Frequency Analysis Using FFT

The Fast Fourier Transform (FFT) was applied to the midspan displacement to extract the dominant vibration frequencies. FFT converts a time-domain signal into its frequency-domain representation, revealing how strongly the structure vibrates at each frequency. The largest spectral peak corresponds to the damped natural frequency, which is the actual frequency at which the cable oscillates when damping is present [8]. Because the damping ratio of the tested system is small, the damped natural frequency lies very close to the undamped natural frequency. Simulation results indicate that the fundamental natural frequency of the cable is approximately 0.40 Hz.

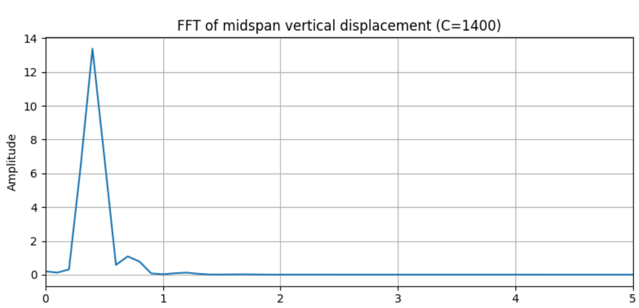

Fig. 13. Frequency spectrum of midspan displacement

This frequency value plays a critical role in engineering evaluation. It identifies the frequency range where the cable is most vulnerable to vibration and determines whether resonance could occur when exposed to periodic excitations such as wind-induced vortex shedding or deck movement. The FFT spectrum also shows several small secondary peaks, which are expected. These minor peaks arise from weakly excited higher-order modes, mild geometric nonlinearities, and aerodynamic contributions from drag and lift. Typical cable mode shapes indicate that the first mode dominates most structural response due to its lower frequency and large midspan motion, explaining why the spectrum is heavily concentrated near the primary peak.

When a sinusoidal deck motion is imposed at a frequency matching the cable's natural frequency, the cable exhibits a strong resonant response characterized by rapidly increasing midspan amplitude. The anchorage point (node 0) is driven by a vertical sinusoidal input with amplitude A=0.02m and frequency f=0.40Hz. When the driving frequency matches the cable's damped natural frequency, the system exhibits a resonant response characterized by rapid growth in vibration amplitude and near-unstable behavior [9].

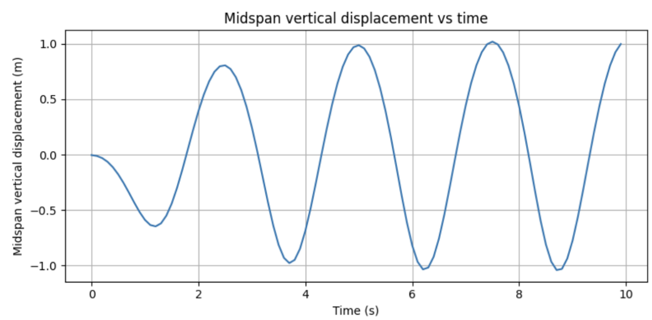

Fig. 14. Midspan displacement under resonant deck excitation

In contrast to the resonant case, Fig. 15 illustrates the cable's midspan response when the deck is excited at a non-resonant frequency of $f = 1$ Hz. Because the external forcing is far from resonance, the cable is unable to efficiently absorb and amplify the input energy. As a result, the vibration amplitude remains small. This behavior confirms that when the excitation frequency does not match the cable's natural frequency, the system avoids the large, potentially unstable oscillations observed under resonant deck excitation.

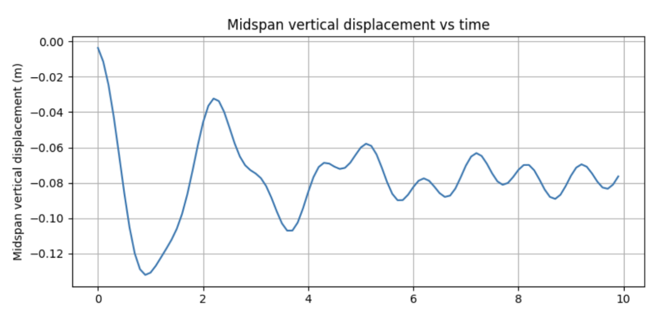

Fig. 15. Midspan displacement under non-resonant deck excitation

V. Future Plan and Vibration Analysis

Future development of this model can proceed in several directions. First, the simulation with external deck excitation may be refined using smaller time steps to capture the resonant growth more accurately. Second, the inclusion of a tuned mass damper could be explored to provide frequency-targeted vibration suppression, particularly under near-resonant loading [5]. Third, a full modal analysis could be conducted to compute higher-mode shapes and modal masses directly, improving the accuracy of damping and frequency predictions. Finally, incorporating more sophisticated aerodynamic models or wind-structure interaction effects would allow the framework to be applied to real bridge environments, where cable vibrations are strongly influenced by wind [10].

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